

Deepwater Horizon oil spill and oyster populations

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Background

An ecological calamity began on April 20, 2010, as crude oil began rapidly leaking from the Deepwater Horizon oil platform into the Gulf of Mexico. The uncontrolled flow of crude oil into the waters of the Gulf of Mexico would not be stopped for 87 days, releasing an estimated 3.19 million barrels (or 134 million gallons) of oil. Almost immediately labeled as one of the worst ecological disasters of our time, scientists knew the damage to ecosystems would be widespread. But because nothing like this magnitude had ever occurred before, no one was really sure quite *how* intense and specifically *which* organisms would be affected.

Hundreds of research projects began, ranging from exploring how fast and deep the oil spread, to the effect on wildlife and ecology. One species of particular interest to researchers was the eastern oyster (*Crassostrea virginica*). In the Gulf, these oysters are a keystone species and extremely important to the ecosystem. Their shells build up important habitats and their activity as filter feeders helps filter the water and allow sunlight to reach the ecosystem. On top of their ecological benefit, the annual oyster harvest in the Gulf of Mexico is worth more than \$60 million annually.

The timeframe of the oil spill aligned perfectly with “a period that coincided with the spawning and recruitment season of eastern oysters in the Gulf of Mexico.” The developing oyster larvae are pelagic or free swimming and spend 2 to 3 weeks in the open sea, generally floating near the surface, until they sink and settle on a suitable surface where they will grow to adulthood. (Vignier et al., 2016). Scientists realized that different types of oil (both surface-collected oil, and oil mixed with chemical oil dispersants under the surface) would likely be toxic to slow moving oyster larvae and hinder growth and therefore the overall population. However, the exact effects of exposure to the oil spill and chemicals used to disperse the oil on oyster larvae remained unknown.

How exactly does exposure to different concentrations of oil contamination affect oysters? Researcher Dr. Julian Vignier of the Laboratoire des Sciences de l'Environnement Marin in France and his team investigated this question, taking data on how different types of oil and concentrations affect settlement success in oyster larvae. Adult oysters were collected from natural populations, kept in hatcheries, and encouraged to spawn. The resulting larvae were then exposed to different oil types and concentrations, and their

settlement rates recorded.

Dataset

This dataset was gathered from a study conducted in response to the Deepwater Horizon Oil spill. It was originally published in a journal called [Marine Environmental Research](#).

Citation:

J. Vignier, P. Soudant, F.L.E. Chu, J.M. Morris, M.W. Carney, C.R. Lay, M.O. Krasnec, R. Robert, A.K. Volety, "Lethal and sub-lethal effects of Deepwater Horizon slick oil and dispersant on oyster (*Crassostrea virginica*) larvae". *Marine Environmental Research*, Volume 120, 2016

Using larvae that were harvested from oyster hatcheries, scientists recreated natural settlement conditions (while also introducing oil concentrations of varying amounts). After 72 hours of exposure, "...settlement plates and container walls were examined under a dissecting microscope, and newly settled oysters counted". With this experiment the scientists were able to compare the oyster larvae settlement rates with different exposure to oil and dispersants.

Each row in the dataset represents the results of one replicate sample of many oyster larvae used in the experiment.

Variables

Treatment - describes the type of oil that the oyster larvae we exposed to.

Concentration - concentration of oil in the water. Measured in milligrams per liter.

Total exposed - number of oyster larvae used in the trial.

Number settled - number of oyster larvae which successfully settled in the experimental container to continue growth and development.

Number unsettled - number of oyster larvae which were unable to successfully settle and could not continue to grow and develop.

Settlement Rate - proportion of settled oyster larvae. It is the number of settled larvae divided by the total number of larvae in a particular trial.

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